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UNIVERSITI TEKNOLOGI MALAYSIA

SAIA 1013: Responsible AI and Ethics

MEETING 4
SECTION 01

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MEETING 4

Risk Assessment for AI Ethics

- Introduction to Risk Assessment
- Fundamental Ethical Risk Assessment and Management of AI Project
- The Ethical Matrix of AI Systems and Data Science

(Some are in form of these activities: Lecture, Discussion, Case Studies, Short Quiz, Self Study)

Lecture Plan Meeting 4

- Introduction to Risk Assessment
- Importance of Risk Assessment in AI
- Types of Risk Assessment Framework
- Steps Involved in Risk Assessment
- Risk Matrix
- Exercise and Reminder of CADR

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Learning Outcome

By the end of this module, students will be able to:

1. Identify and assess ethical risks in AI projects using a risk assessment approach.

2. Apply the Ethical Matrix to evaluate decisions in AI systems and data science.

“Threats are Embraced not Denied,

Problems are Identified not Hidden”

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INTRODUCTION TO RISK ASSESSMENT

- A **risk assessment (RA)**, also referred to as "risk analysis," is a process used to identify and evaluate risks. Risks are then quantified based on their **importance or impact severity**. These risks are then **prioritized.**
- RA is used to help identify which safeguards to implement. **Safeguards** are also known as controls.



Risk Assessment is created at a moment in time

Risk Management program is a continuous process.



"Risk management refers to coordinated activities to direct and control an organization with regard to risk" (Source: ISO 31000:2018).

https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf

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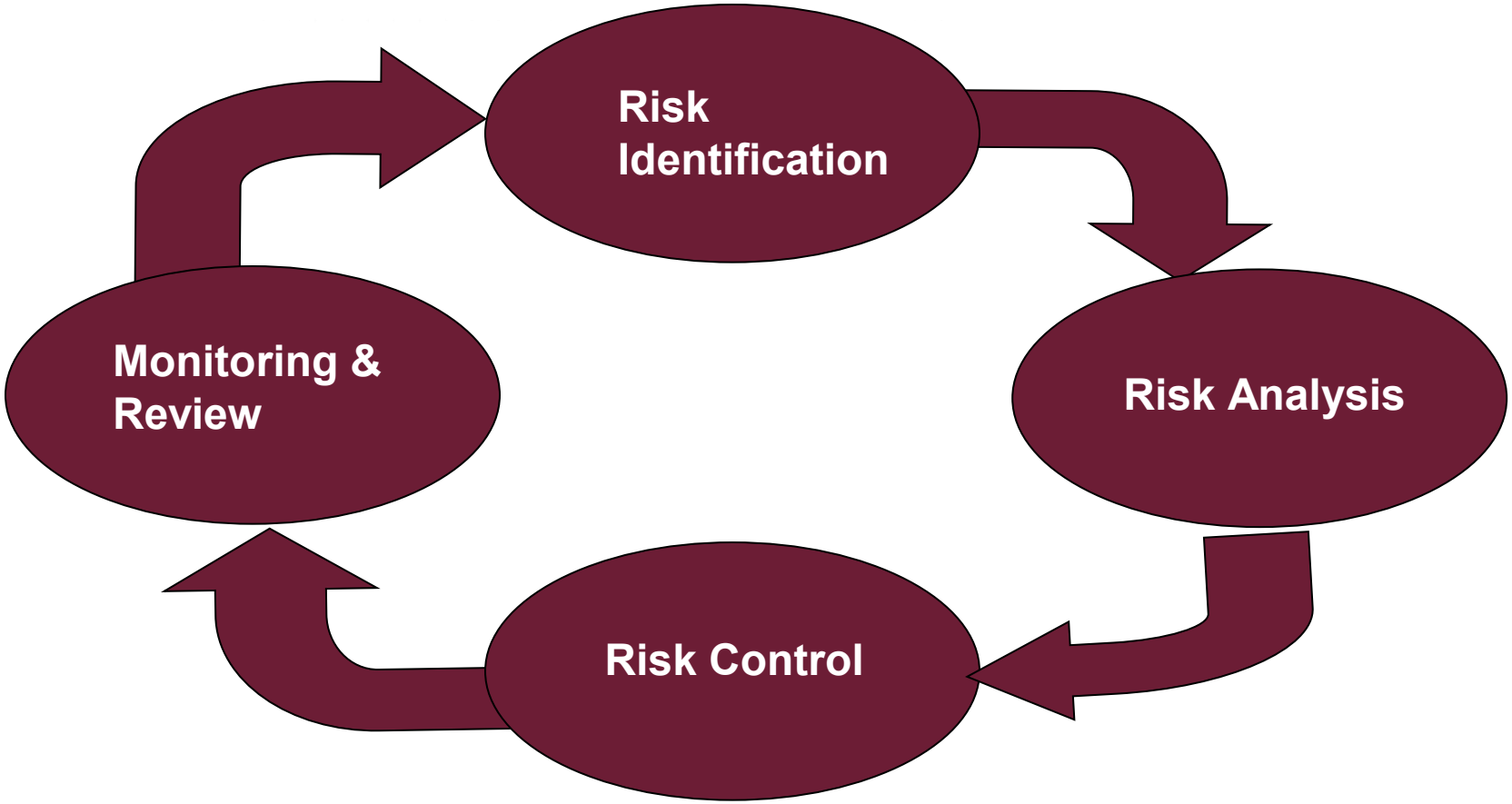
Overview

The Risk Management Process consists of a series of steps that, when undertaken in sequence, enable continual improvement in decision-making.

Risk can come in different ways:

- Schedule risks (short to long)
- Cost risks (small to large)
- Technology risks (probable to impossible)

The General Risk Management Cycle



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LET US TRY SIMPLE PROCESS OF RISK MANAGEMENT

1.RISK IDENTIFICATION

Let's Perform Risk Identification On Your Own Home Computer.

1. Hard drive failure
2. Spyware
3. Malware
4. Theft
5. Other hardware failure
6. Loss of internet connectivity

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Con't..

2. RISK ASSESSMENT/PRIORITIZATION

Risk	Probability/Likehood	Risk Impact/Severity
Hard drive failure	Medium	High
Spyware	Medium	Low
Malware	Low	High
Theft	Low	High
Other hardware Failure	Low	Medium/High
Loss of Internet connectivity	Low	Low

Note: must provide reasoning for the rating.

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Con't..

3. RISK MITIGATION

Aim: to make less severe or reduce its impact

Example:

✓What action can you do to mitigate the risk of spyware or malware on our home computer

✓Solution :

- Install anti –virus
  - Hardware firewall or anti-virus?
- Perform cost benefit analysis…
  - How much you willing to spend???

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Con't..

4. RISK MONITORING/CONTROL

- Remember all the risk can never be eliminated.
- A procedure must be put in place
- Update anti-virus
- Regularly create-backups
- Install software in a sandbox environment such as Virtual Machine

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The Ethical Risk Assessment (e.g. For AI Project)

1. Risk Identification:
Goal: Spot potential ethical risks early in the AI development lifecycle.
AI Context Example:

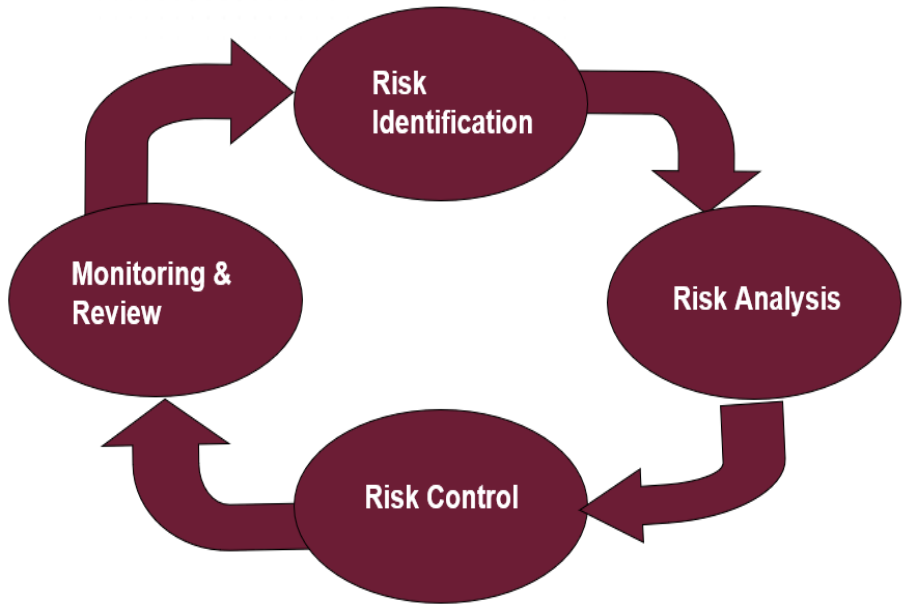
- Identifying risks like bias in training data, lack of transparency, or privacy violations.
- Example: A facial recognition system may misidentify individuals from certain ethnic groups due to biased datasets.

2. Risk Analysis
Goal: Evaluate the severity and likelihood of each identified risk.
AI Context Example:

- Assessing how likely it is that an AI chatbot might generate harmful or misleading content.
- Example: Analyzing the impact of a recommendation algorithm reinforcing echo chambers or misinformation.

3. Risk Control
Goal: Implement measures to mitigate or eliminate risks.
AI Context Example:

- Applying fairness constraints, data anonymization, or human-in-the-loop systems.
- Example: Introducing explainability tools to help users understand how an AI model make decision.



3. Monitoring & Review
Goal: Continuously track performance and ethical compliance post-deployment.
AI Context Example:

- Regular audits of model outputs, retraining with updated data, and feedback loops.
- Example: Monitoring a predictive policing tool to ensure it doesn't disproportionately target specific communities over time.

When to Perform Risk Management

(WORST) *Crisis management/Fire fighting* : addressing risk after they present a big problem

(BAD) *Fix on failure* : finding and addressing as they occur

(OKAY) *Risk Mitigation* : plan ahead and allocate resources to address risk that occur, but don't try to eliminate them before they occur

(GOOD) *Prevention* : part of the plan to identify and prevent risks before they become problems

(BEST) *Eliminate Root Causes* : part of the plan to identify and eliminate the factors that make specific risks possible

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Critical components of risk assessment

1. Identify Scope

How do you choose? Who can make decision?

2. Identify critical areas

From scope, identify critical areas

3. Identify team

Risk assessment team personnel should not be the same people who are responsible for correcting deficiencies. This helps avoid a conflict of interest.

For example:

Identify Scope (Decision Authority)

Example: Deciding whether to use **AI proctoring software** for online exams.

Question: Who makes the decision — faculty, IT services, or university senate?

Identify Critical Areas (Risk Focus)

Example: From AI proctoring scope, critical risks include:

Privacy → webcam monitoring of students at home.

Fairness → software may misidentify students with certain skin tones or disabilities.

Reliability → false cheating alerts could harm students' academic records.

Identify Team (Independence & Expertise)

Example: Risk assessment team composition:

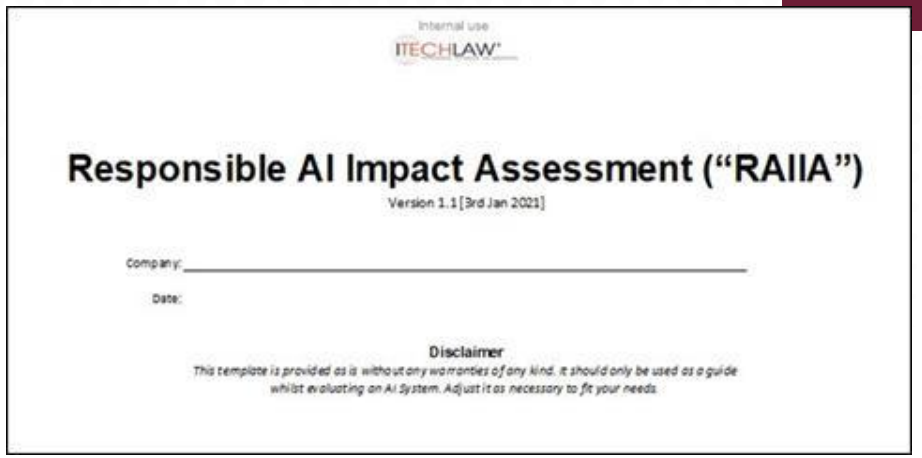
Academic staff representative (to check fairness in grading).

Student representative (to voice student rights & concerns).

Data security officer (to ensure student data is handled ethically).

Importantly: **team is separate** from the IT department implementing the system.

Example Documentation of the RAIIA Used for Responsible AI Impact Assessment



Internal use

ITECHLAW

Responsible AI Impact Assessment ("RAIIA")

Version 1.1 [3rd Jan 2021]

Company: _____

Date:

Disclaimer

This template is provided as is without any warranties of any kind. It should only be used as a guide whilst evaluating an AI System. Adjust it as necessary to fit your needs.

Example of Components

1. Project Summary
2. Key factors for Conducting RAIIA
3. Main Assessment
   a. Principle 1: Ethical Purpose and Societal Benefits
   b. Accountability
   c. Transparency and Explainability
   d. Fairness and Non Discrimination
   e. Safety and reliability
   f. Open Data , Fair Competition and Intellectual Property
   g. Intellectual Property
   h. Privacy
4. Risk Assessment Summary

	Factors to Evaluate Need for RAIIA		
Context	1. Describe the context in which the AI System will be used or deployed.	Human Oversight	14. Will the AI System make or participate in making decisions with material impacts on individuals or society?
	2. Will the use of the AI System be citizen-facing?		15. What is the expected degree of autonomy of the AI System? Will, for instance, human operators or decision-makers have oversight on individual AI decisions, if any?
	3. What is the market, industry or sector targeted?		16. How frequently will there be human oversight over the operation of the AI System?
Laws and regulations	4. Do the jurisdiction(s) in which the AI Solution will be deployed have data protection laws or regulation that are applicable to its use?		17. What measures would be taken to avoid automation bias or anchoring to the AI System?
	5. Does the jurisdiction(s) in which the Project will take place abide by rule of law principles?		18. What will be the Organisation's degree of control and responsibility over the finalized AI System?
	6. Does this jurisdiction have antidiscrimination laws?	Data and Privacy	19. What is the type and origin of the data that will be used to train the AI System?
	7. What are the main regulatory requirements relevant to the use and deployment of the AI System within the targeted market, industry or sector?		20. Will the training data include personal information?
	8. Will the AI System be used across legal jurisdiction borders (whether they be across federal states or national borders)?		21. If personal information are used in the context of the AI System, who are the data subjects?
	9. What are the main ethical concerns relevant to the use and deployment of the AI System for the targeted market, industry or sector?		22. What is the level of sensitivity of the data in term of privacy?
Stakeholders	10. Who will be the main stakeholders affected by the AI System?	Human-understandable AI	23. What are the technical characteristics of the AI System that could influence the explainability and auditability of the algorithm?
	11. Who are the expected contributing third parties?		24. Can the results of the AI System be explained in humanly understandable terms?
	12. What individual rights and interests will be at stake as a consequence of the use of the AI System?		
	13. Are those rights fundamental or human rights?		

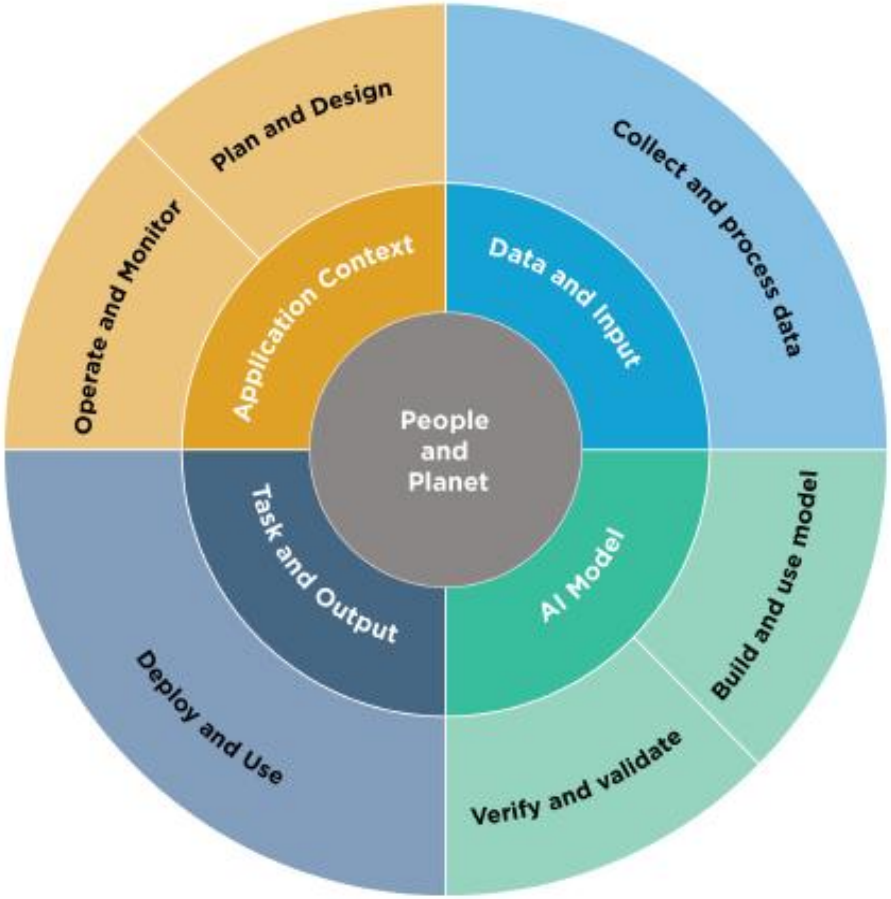
Examples of potential harms related to AI systems.



Trustworthy AI systems and their responsible use can mitigate negative risks and contribute to benefits for people, organizations, and ecosystems.

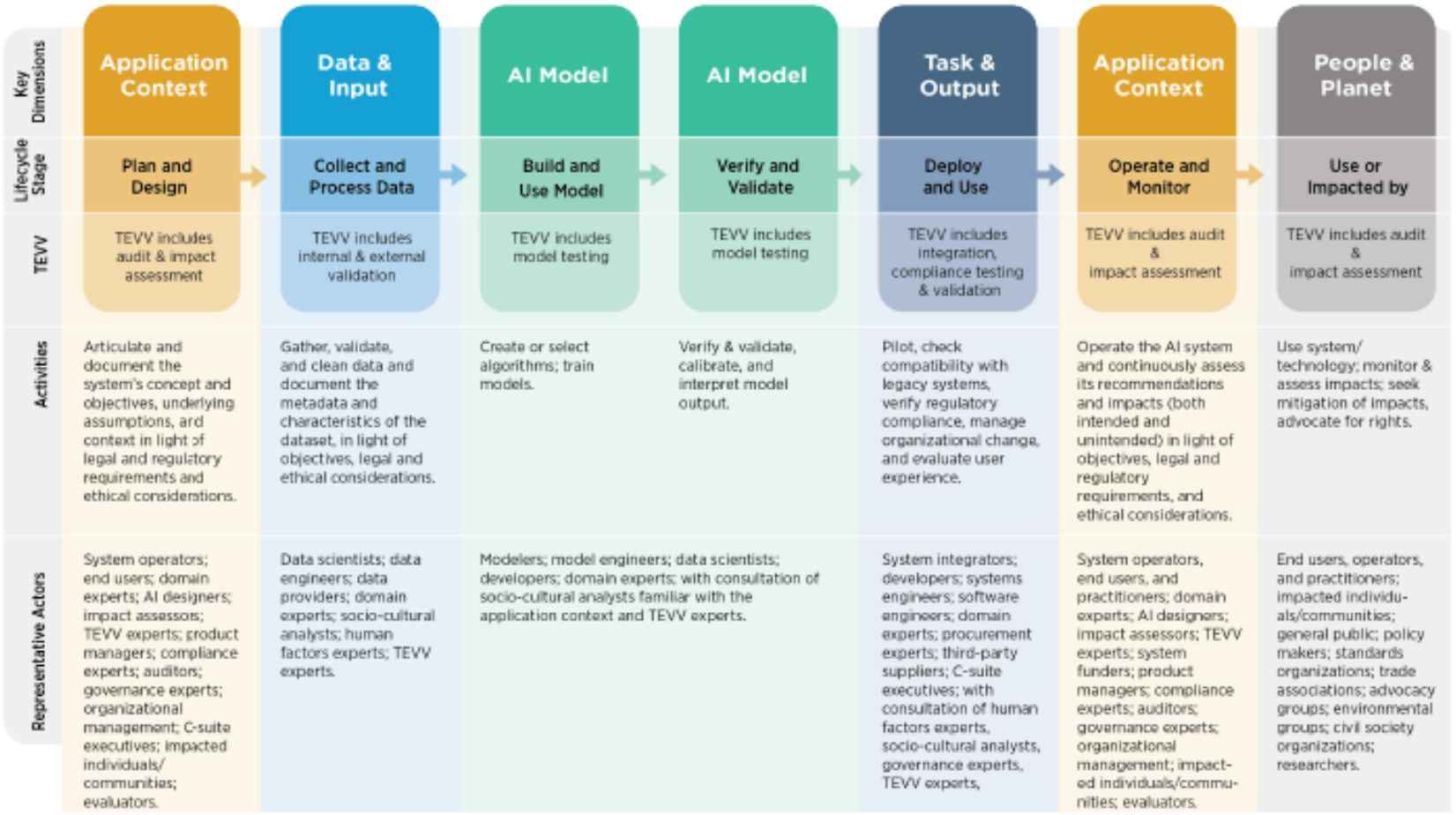
Lifecycle and Key Dimensions of an AI System.

Modified from OECD (2022)



- ✓ The two inner circles show AI systems' key dimensions, and the outer circle shows AI lifecycle stages.
- ✓ Ideally, risk management efforts start with the Plan and Design function in the application context and are performed throughout the AI system lifecycle.

AI lifecycle stages



Key Dimensions	Application Context	Data & Input	AI Model	AI Model	Task & Output	Application Context	People & Planet
Lifecycle Stage	Plan and Design	Collect and Process Data	Build and Use Model	Verify and Validate	Deploy and Use	Operate and Monitor	Use or Impacted by
TEVV	TEVV includes audit & impact assessment	TEVV includes internal & external validation	TEVV includes model testing	TEVV includes model testing	TEVV includes integration, compliance testing & validation	TEVV includes audit & impact assessment	TEVV includes audit & impact assessment
Activities	Articulate and document the system's concept and objectives, underlying assumptions, and context in light of legal and regulatory requirements and ethical considerations.	Gather, validate, and clean data and document the metadata and characteristics of the dataset, in light of objectives, legal and ethical considerations.	Create or select algorithms; train models.	Verify & validate, calibrate, and interpret model output.	Pilot, check compatibility with legacy systems, verify regulatory compliance, manage organizational change, and evaluate user experience.	Operate the AI system and continuously assess its recommendations and impacts (both intended and unintended) in light of objectives, legal and regulatory requirements, and ethical considerations.	Use system/ technology; monitor & assess impacts; seek mitigation of impacts, advocate for rights.
Representative Actors	System operators; end users; domain experts; AI designers; impact assessors; TEVV experts; product managers; compliance experts; auditors; governance experts; organizational management; C-suite executives; impacted individuals/ communities; evaluators.	Data scientists; data engineers; data providers; domain experts; socio-cultural analysts; human factors experts; TEVV experts.	Modelers; model engineers; data scientists; developers; domain experts; with consultation of socio-cultural analysts familiar with the application context and TEVV experts.		System integrators; developers; systems engineers; software engineers; domain experts; procurement experts; third-party suppliers; C-suite executives; with consultation of human factors experts, socio-cultural analysts, governance experts, TEVV experts,	System operators, end users, and practitioners; domain experts; AI designers; impact assessors; TEVV experts; system funders; product managers; compliance experts; auditors; governance experts; organizational management; impact-ed individuals/commu-nities; evaluators.	End users, operators, and practitioners; impacted individu-als/communities; general public; policy makers; standards organizations; trade associations; advocacy groups; environmental groups; civil society organizations; researchers.

- A detailed descriptions of AI actor tasks, including details about testing,
- evaluation, verification, and validation tasks

SEVERAL FRAMEWORK USED IN RISK ASSESSMENT

National Institute of Standards and Technology (NIST) Risk Management Framework

According to AI RMF – Definition of Risk

- Risk = Probability of an event × Magnitude of its consequences
- Impacts can be:
  - Positive → opportunities, benefits
  - Negative → threats, harms
- Source: Adapted from ISO 31000:2018

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AI Risk Management Framework Core (NIST AI 100-1)



- Core is composed of four functions: GOVERN, MAP, MEASURE, and MANAGE.
- Each of these high-level functions is broken down into categories and subcategories.
- Categories and subcategories are subdivided into specific actions and outcomes.
- Actions do not constitute a checklist, nor are they necessarily an ordered set of steps.
- Risk management should be continuous, timely, and performed throughout the AI system lifecycle dimensions.

Further reading: Read Page 20 – 33

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AI Risk Management Framework (NIST AI 100-1)

Framework users may apply these functions as best suits their needs for managing AI risks based on their resources and capabilities. Some organizations may choose to select from among the categories and subcategories; others may choose and have the capacity to apply all categories and subcategories. Assuming a governance structure is in place, functions may be performed in any order across the AI lifecycle as deemed to add value by a user of the framework. After instituting the outcomes in **GOVERN**, most users of the AI RMF would start with the **MAP** function and continue to **MEASURE** or **MANAGE**. However users integrate the functions, the process should be iterative, with cross-referencing between functions as necessary. Similarly, there are categories and subcategories with elements that apply to multiple functions, or that logically should take place before certain subcategory decisions.

GOVERN (NIST AI 100-1)

- GOVERN is a cross-cutting function that is infused throughout AI risk management and enables the other functions of the process. Aspects of GOVERN, especially those related to compliance or evaluation, should be integrated into each of the other functions.
- Attention to governance is a continual and intrinsic requirement for effective AI risk management over an AI system's lifespan and the organization's hierarchy.

MAP (NIST AI 100-1)

- The MAP function establishes the context to frame risks related to an AI system.
- The AI lifecycle consists of many interdependent activities involving a diverse set stakeholders
- In practice, AI stakeholders in charge of one part of the process often do not have full visibility or control over other parts and their associated contexts.
- The interdependencies between these activities, and among the relevant AI actors, can make it difficult to reliably anticipate impacts of AI systems. For example, early decisions in identifying purposes and objectives of an AI system can alter its behavior and capabilities, and the dynamics of deployment setting (such as end users or impacted individuals) can shape the impacts of AI system decisions.
- As a result, the best intentions within one dimension of the AI lifecycle can be undermined via interactions with decisions and conditions in other, later activities.

MEASURE (NIST AI 100-1)

- The MEASURE function employs quantitative, qualitative, or mixed-method tools, techniques, and methodologies to analyze, assess, benchmark, and monitor AI risk and related impacts. It uses knowledge relevant to AI risks identified in the MAP function and informs the MANAGE function.
- AI systems should be tested before their deployment and regularly while in operation. AI risk measurements include documenting aspects of systems' functionality and trustworthiness.

MANAGE (NIST AI 100-1)

- The MANAGE function entails allocating risk resources to mapp and measure risks on a regular basis and as defined by the GOVERN function.
- Risk treatment comprises plans to respond to, recover from, and communicate about incidents or events.

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Risk Management Process



Ref. *Guide for the Process of Managing Risk on Rapid Renewal Projects* 3

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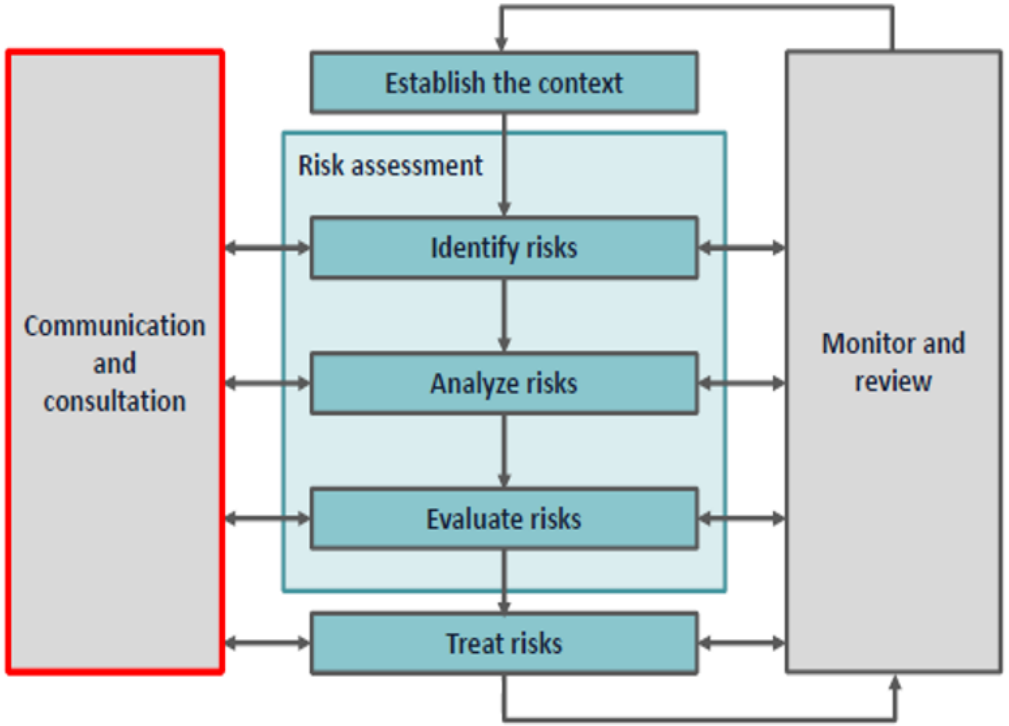
The AUS/NZ Risk Management Process

- Establish the context
- Identify risks
- Analyse
- Evaluate
- Treat
- Communicate
- Monitor and Review

(NIST AI 100-1)



RISK MANAGEMENT ISO 31000 PROCESS

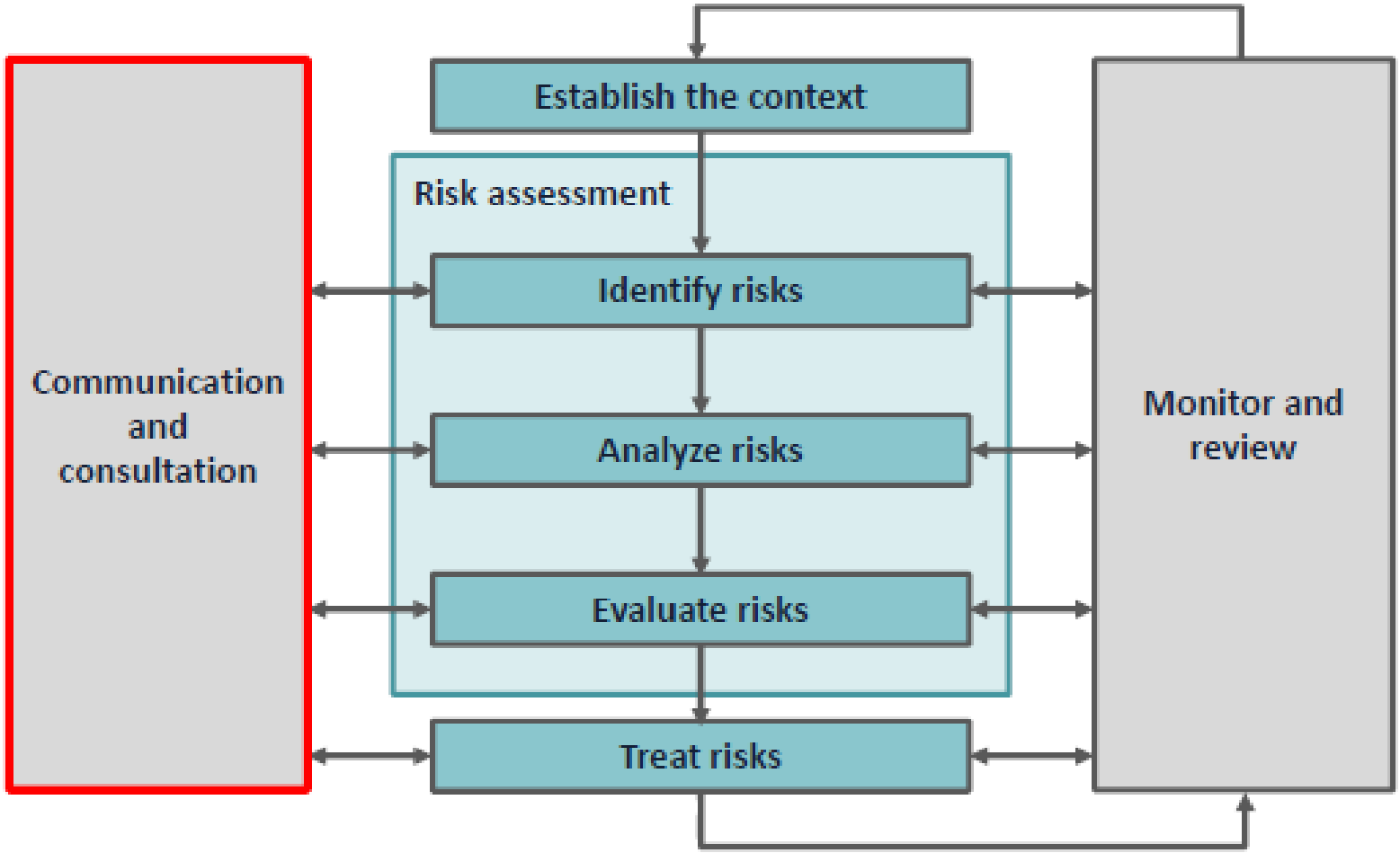


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RISK MANAGEMENT
ISO 31000 PROCESS

Simply just focus on:

1. Identify
2. Analyze
3. Evaluate
4. Treat



MAPPING ISO 31000 VS AUS/NZ Risk Management Process VS NIST AI 100-1

Risk Assessment Column	ISO 31000 & AUS/NZ Risk Management Process	NIST AI 100-1
Risk Type	Establish the context	Map
AI Principle Violated	Identify risks	Measure
Ethical Risk Identified	Analyse risks	Measure
Affected Stakeholder(s)	Establish the context / Communicate	Map / Govern
Intensity of Risk	Analyse risks	Measure
Likelihood to Happen	Analyze risks	Measure
Risk Category	Establish the context	Map
Proposed Mitigation	Treat risks	Manage

Template for Risk Assessment

Column	Description
Risk Type	Depending on the risk being registered
AI Principle Violated	Identify which principle is affected (e.g., Fairness, Privacy, Accountability, Transparency, Inclusiveness)
Ethical Risk Identified	Name and briefly describe the ethical risk, e.g.: bias risks, black-box risks, legal/liability risks, surveillance risks, autonomy & control risks, technical robustness risks, etc
Affected Stakeholder(s)	List the individuals or groups impacted
Intensity of Risk	Estimate how severe the impact would be (Low/Medium/High)
Likelihood to Happen	Estimate how likely it is to occur (Rare to Almost Certain), 3x3 or 5x5
Risk Category	Classify the risk (Technical, Safety & Security, Governance, Legal, etc.)
Proposed Mitigation	Suggest concrete ways to reduce or prevent the risk including suggesting with reasoning Risk Treatment Strategy (treat, tolerate, transfer, terminate)

Common Types of Risk Assessment

Quantitative

This is an objective method. It uses numbers such as actual dollar values. A quantitative RA requires a significant amount of data. Gathering this data often takes time. If the data is available, this type of RA becomes a simple math problem with the use of formulas.

Qualitative

This is a subjective method. It uses relative values based on opinions from experts. Experts provide their input on the probability and impact of a risk. A qualitative RA can be completed rather quickly.

Can also be mixed.

Probability (likelihood) and Impact (Intensity)

TABLE 5-1 Probability scale.

PROBABILITY	DESCRIPTION
Low	It is unlikely the risk will occur. Threats are not active. Vulnerabilities are either not known or have been mitigated. Low equates to a value of 10 percent.
Medium	There is a moderate chance the risk will occur. It has occurred in the past, but mitigation controls have reduced recent occurrences. Medium equates to a value of 50 percent.
High	There is a high probability the risk will occur. It has occurred in the past and will occur again if not mitigated. High equates to a value of 100 percent.

TABLE 5-2 Impact scale.

IMPACT	DESCRIPTION
Low	If the risk occurs, it will have minimal impact on the company. The attack will not impact any critical data or systems.
Medium	If the risk occurs, it will have a moderate impact on the company. It may affect critical data or systems, but not to a large extent.
High	If the risk occurs, it will have a high impact on the company. It will affect critical data or systems and cause substantial losses.

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Quantitative risk assessment

1. Risk Scores

A numerical value that represents the level of risk, calculated by multiplying **Likelihood × Impact**.

Purpose: Provides a quick way to compare risks and prioritize them.

Example (AI in education):

- Likelihood of AI grading system error = 0.3 (30%)
- Impact on student outcomes = 4 (High, on 1–5 scale)
- Risk Score = 0.3 × 4 = 1.2 → Medium priority risk.

2. Annualized Loss Expectancy (ALE)

A financial measure of risk, estimating the expected yearly cost of a risk event.

Formula: ALE = *Single Loss Expectancy (SLE)* × *Annual Rate of Occurrence (ARO)*

- Purpose: Helps organizations budget for risk mitigation by translating risks into monetary values.
- Example (AI system data breach at a university):
  - SLE = RM 200,000 (cost of one breach, including fines & reputational loss)
  - ARO = 0.2 (once every 5 years)
  - ALE = RM 40,000 per year.

Quantitative risk assessment

3. Safeguard Value (Cost of Controls)

The cost of implementing a control to reduce or eliminate risk.

AI Example: University introduces a human-in-the-loop review system to check AI proctoring flags before action is taken.

Cost of safeguard = $40,000 per year, If safeguard reduces risk by 80%, expected benefit = $120,000 avoided loss

Net gain = $80,000 (worth implementing).

Note: Must gather information from Human Resource or Other Affecting Department.

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Qualitative risk assessment

- A **qualitative risk assessment** doesn't assign dollar values. Instead, it determines the level of risk based on the **probability** and **impact** of a risk. You determine these values by ***gathering the opinions of experts. (Delphi Method)***
- An impact assessment risk level could use words such as **Low, Medium, or High**.
- **Probability**—The likelihood that a threat will exploit a vulnerability. The risk occurs when a threat exploits a vulnerability. You can use a scale to de ne the probability that a risk will occur. The scale can be based on word values such as Low, Medium, or High.
- **Impact**—The negative result if a risk occurs. Impact is used to identify the magnitude of a risk. The risk results in some type of loss. However, instead of quantifying the loss as a dollar amount.

RISK FORMULA :

Risk level = Probability (or Likelihood) x Impact

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Qualitative risk assessment

Risk Matrices

A **visual tool** (usually a grid) that categorizes risks based on their **likelihood** (y-axis) and **impact** (x-axis).

Purpose: Simplifies decision-making by showing which risks are Low, Medium, High, or Critical

Example (AI exam proctoring tool):

✓Likelihood: High (frequent false positives)

✓Impact: High (students unfairly penalized)

✓Risk Level: **Critical (Red Zone)** on the matrix.

What is risk analysis?

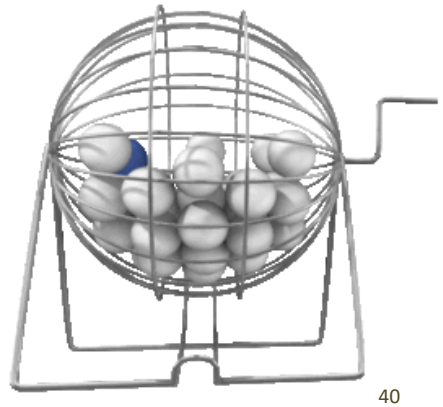
- **A stage in the risk management process where risks are rated according to the criteria established by the organization**
- **The most common criteria used for risk rating are the probability/likelihood and the impact**
- **Allows for risk profiling, using risk matrices**
- **This allows managers to prioritize risks for the urgency and degree of treatment**
  - Separate **major risks** from **minor risks**
  - See which risks need **regular checks**
  - Identify **suitable risk responses** and
  - Identify **suitable risk reports and assurance**



Dimensions of risk analysis

- **Likelihood (qualitative)**
  - **Estimates** the frequency or probability of an event in **qualitative terms**
  - Example: High, medium or low likelihood

- **Probability (quantitative)**
  - **Calculates** the extent to which an event is likely to occur, expressed in quantitative terms as a value between 0% and 100%
  - Example: 70% chance of rain on any day



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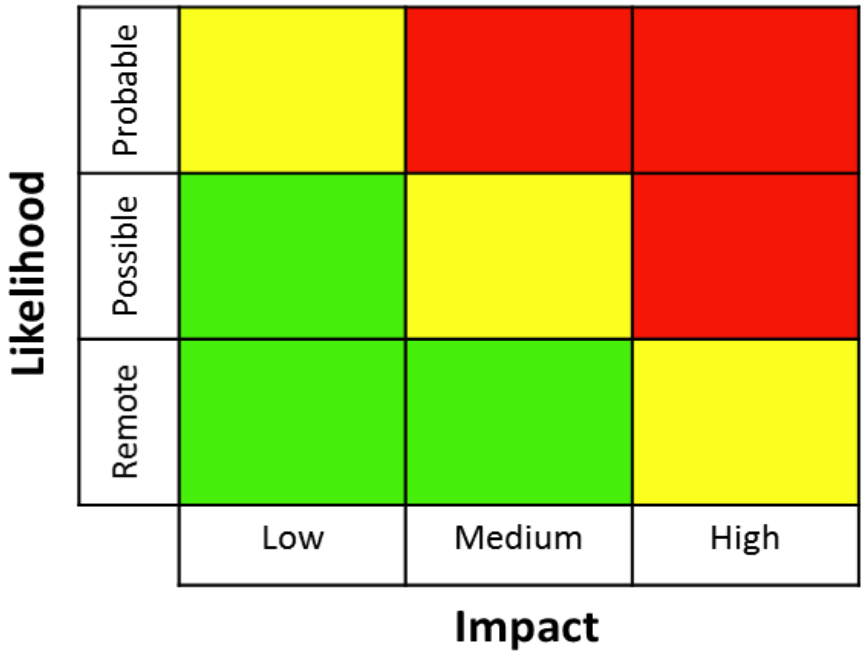
Dimensions of risk analysis

- **Impact**
  - Represents the outcome of an event measured **quantitatively** or **qualitatively**
  - Example: A measurement of financial loss: Low impact (<$50K), Medium impact ($50K - $200K), High impact (>$200K)



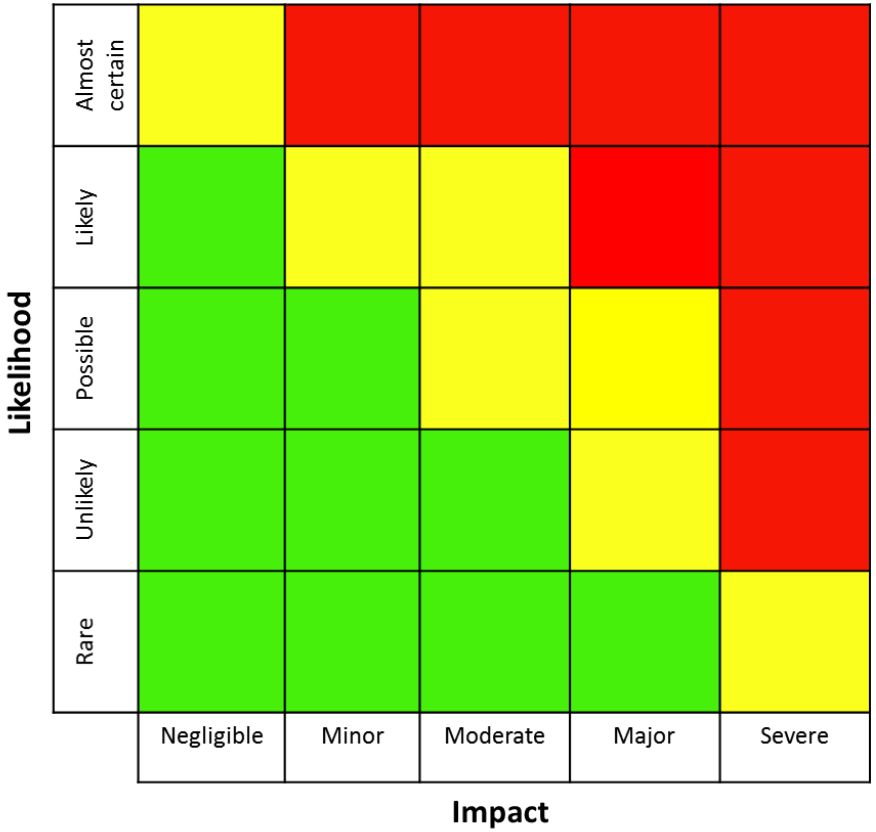
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Risk matrix – qualitative analysis



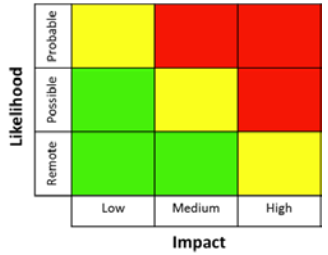
Meaning of colors:

- Low risks
- Intermediate risks
- Critical risks

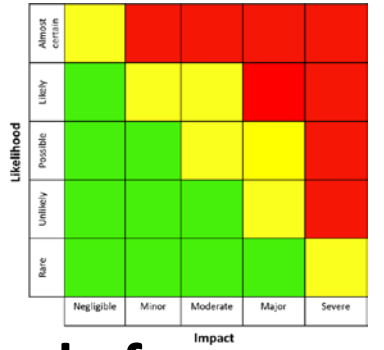


Rating impact and likelihood

- **Many organizations use a simple 3x3 matrix**



- **Other organizations prefer a 5x5 matrix in order to have a more refined risk rating**



- **Each organization should define the needed level of accuracy according to its risk management strategy**

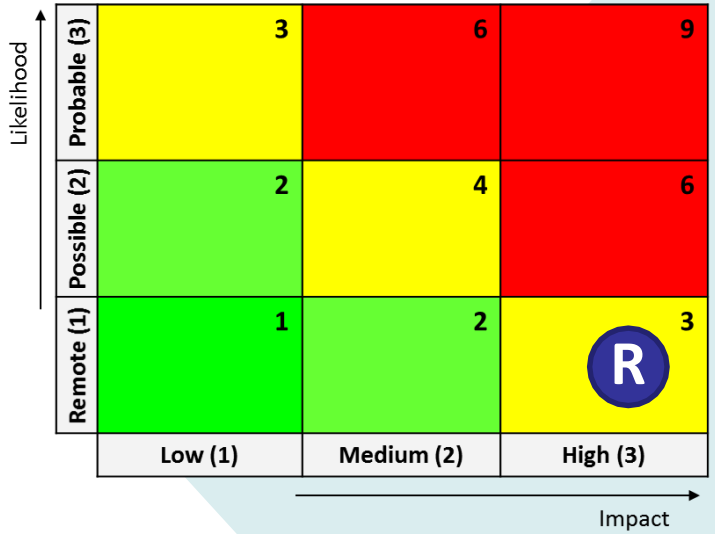
- **Each level of likelihood and impact can be given a number, in order to represent the RISK EXPOSURE**

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Risk exposure

- **Example with a (3 x 3) Risk Matrix**



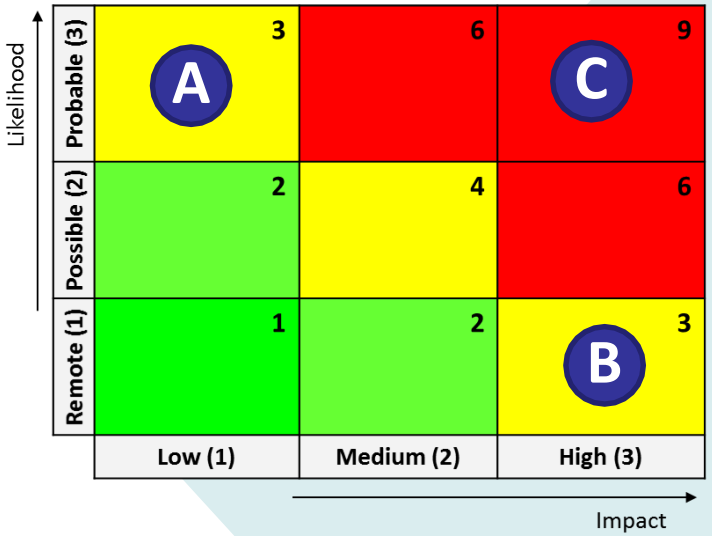
- We assign a number to each level of likelihood and impact. In this example the numbers 1, 2 and 3 are used.
- The number in each cell represents a risk exposure: **Likelihood x Impact**
- A risk R with the following profile:
  - Likelihood = Remote (1)
  - Impact = High (3)
- has an exposure of **3** (1 x 3) and will be in the lower right cell.

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Risk exposure

- **Example with a 3x3 Risk Matrix**



Now suppose we have 3 risks – **A**, **B** and **C** – with the profiles represented in the risk matrix on the left.

The Risk Exposures are the following:

- Risk A = 3 (3 x 1)
- Risk B = 3 (1 x 3)
- Risk C = 9 (3 x 3)

The total risk exposure will be:

(3 + 3 + 9) / 3 = **5**

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What is risk evaluation?

- **It is the last stage of risk assessment. It is the ‘so what?’ stage**
- **The purpose is to assist in making decisions, based on the outcomes of risk analysis, about which risks need treatment and the priority for treatment implementation**
- **Involves comparing the level of risk found during the analysis process with risk criteria established when the context was considered**
- **Based on this comparison, the need for treatment can be considered**

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Definitions around risk evaluation

Risk appetite	The amount of risk an organization is willing to seek or accept in pursuit of its long-term objectives.
Risk tolerance	Acceptable level of variation in the risk level an organization is willing to accept regarding the pursuit of its objectives.
Risk universe	The full range of risks which could impact, either positively or negatively, on the ability of the organization to achieve its long-term objectives.
Risk capacity	The capability of the organization to take risk, without going bankruptcy.

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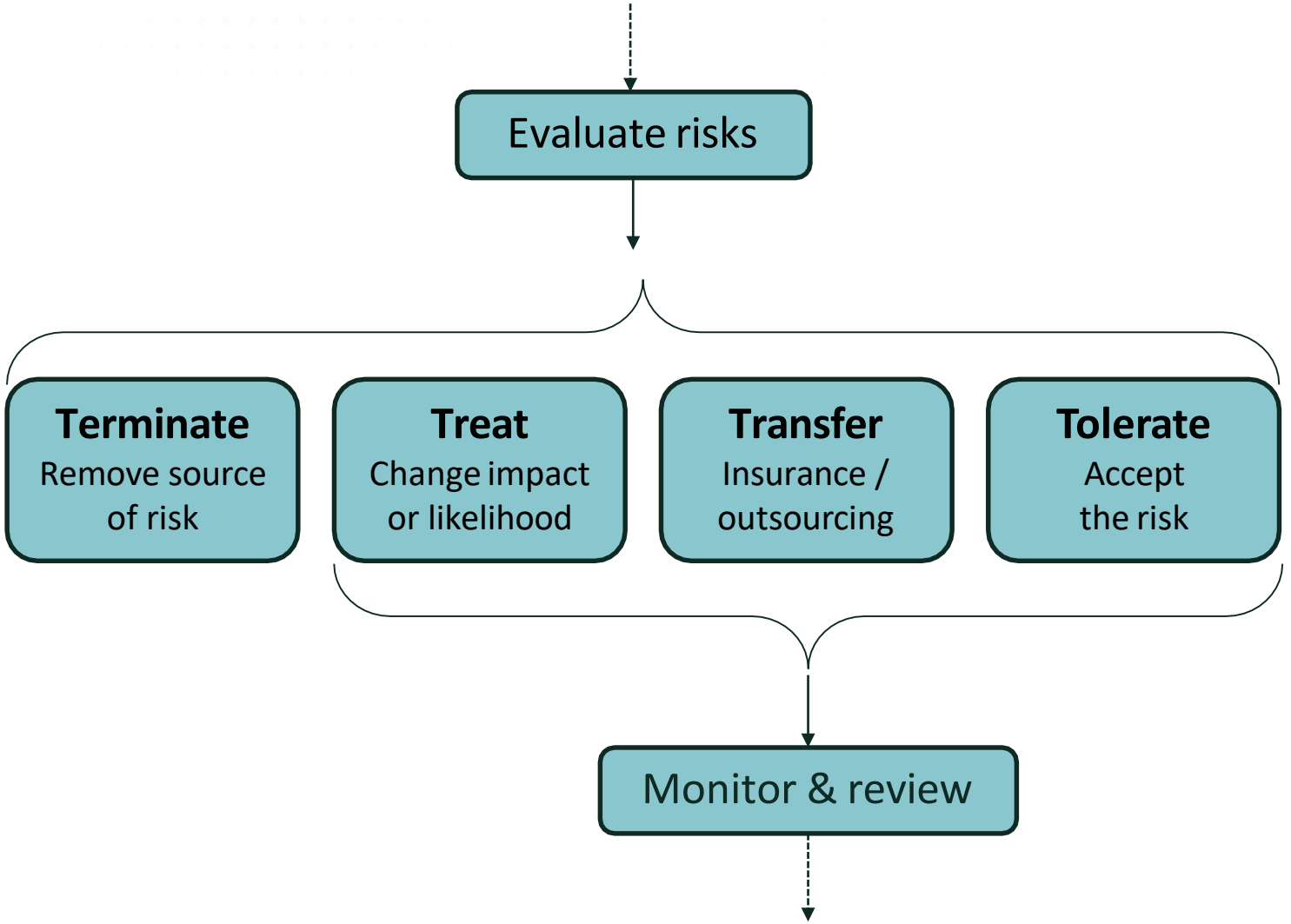
What is risk treatment?

- **A process to modify the risk profile**
- **Risk treatment (or response) involves:**
  - The selection of one or more options for modifying risks
  - Implementing those options
  - The treatments then provide controls or modify current controls

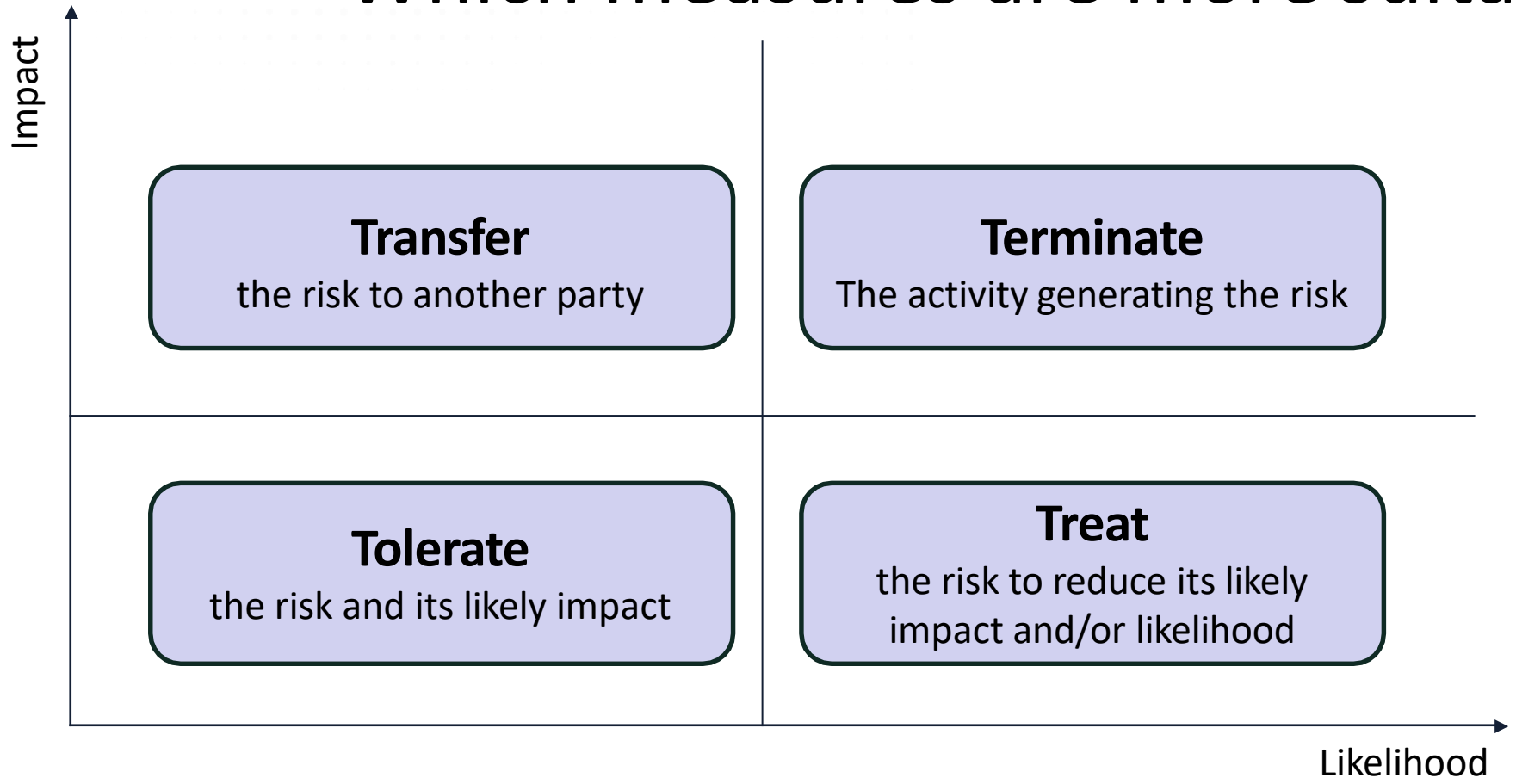


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Risk response options



Which measures are more suitable?



Source: Paul Hopkin (2014). Fundamentals of Risk Management, Kogan Page, London.

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Exercise

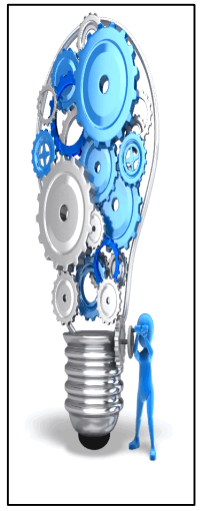


Which treatment measure is implied in each of the following actions? What is the reasoning?

1. Bias audit to be conducted by an external AI ethics consultancy.
2. Shutting down an AI application that consistently produces harmful outputs.
3. Conducting fairness and transparency training for AI developers.
4. Delaying the appointment of an AI Ethics Lead for one year.

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Answers to the exercise



Which treatment measure is implied in the following actions? Reasoning?

1. Transfer
2. Terminate
3. Treat
4. Tolerate

1. Bias audit to be conducted by an external AI ethics consultancy.
Transfer – Outsourcing the responsibility to manage the risk.
2. Shutting down an AI application that consistently produces harmful outputs.
Terminate – Eliminating the risk by removing the source.
3. Conducting fairness and transparency training for AI developers.
Treat – Addressing the risk through education and process improvement.
4. Delaying the appointment of an AI Ethics Lead for one year.
Tolerate – Accepting the risk without immediate action.

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Risk Management's Benefits

- ✓ A proactive rather than reactive approach.
- ✓ Reduces surprises and negative consequences.
- ✓ Prepares the project manager to take advantage of appropriate risks.
  - ➢ Take action when a time, cost and/or technical advantage is possible (exploit resources)
- ✓ Provides better control over the future.
  - ➢ Know what to do when things happen
- ✓ Improves chances of reaching project performance objectives within budget and on time.

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CADR (25% Report; 10% Video Presentation)

- ✓ Refer to the instruction guideline as provided in the e-learning.
- ✓ Use all the knowledge, processes, steps of Risk Assessment provided in this teaching material to help guide and complete your CADR Project requirement.
- ✓ Both (a) report and (b) recording presentation video must be submitted to e-learning on Friday, 12th Sept 2025, Week 6.